

Networking Core Protocols Summary

FTP (File Transfer Protocol)

USER is used to input the username.

PASS is used to enter the password.

RETR (retrieve) is used to download a file from the FTP server to the client.

STOR (store) is used to upload a file from the client to the FTP server.

FTP server listens on TCP port 21 by default; data transfer is conducted via another connection from the client to the server.

Example: ftp 10.10.33.105 to connect to a remote FTP server.

SMTP (Simple Mail Transfer Protocol)

Defines how a mail client talks with a mail server and how servers communicate.

Analogy: like sending a package at a post office.

Commands:

- HELO/EHLO: initiates an SMTP session.
- MAIL FROM: specifies sender's email.
- RCPT TO: specifies recipient's email.
- DATA: begins the message body.
- . (dot on its own line): ends the message.

SMTP server listens on TCP port 25.

POP3 (Post Office Protocol v3)

Used for retrieving emails on a single device.

Commands:

- USER : identifies the user.
- PASS : provides password.
- STAT: requests message count and size.
- LIST: lists all messages.
- RETR : retrieves a message.
- DELE : marks message for deletion.
- QUIT: ends session and applies changes.

POP3 is suited for one-device access; emails are often deleted after download.

IMAP (Internet Message Access Protocol)

Designed for multi-device email synchronization.

Unlike POP3, messages remain on the server and are synchronized across devices.

Commands:

- LOGIN : authenticates user.
- SELECT : selects a folder.
- FETCH : retrieves a message.
- MOVE : moves messages.
- COPY : copies messages.
- LOGOUT: ends session.

IMAP server listens on TCP port 143.